STRhub Verified

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Technical Reproducibility Validation Report

HipSTR

Version v0.7

Autosomal STR

Verification date	2026-11-06
Repository commit	b2033bfbb5cf55496b776463bdf2993fa763a4be
Result	**Runs as documented**
Reached	Runs + Plausible output
Permalink	https://strhub.app/verified/hipstr
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution. It confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**Runs as documented**
REACHED	Runs + Plausible output, 6/6 gates passed
WHY	The tool installed and its run produced its documented output.
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Step 1: The Documented Run

As documented, it runs.

A third party installed the tool with the steps this commit states and ran the command it writes, and the output file that command names appeared. This says nothing about genotypes or about the forensic panel.

3. As it is in the repository: Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Starts	The installed program answers --help in the built environment	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS
Plausible Output	Output structure matches expected genotype-bearing format	PASS
Execution log	hipstr.log-external.txt	

4. Exact Run Command

This is the command that was executed, verbatim, in the CI environment.

```
# Environment: ubuntu-22.04 · HipSTR v0.7 · commit b2033bf
$ cd '/opt/tool' &&
touch /tmp/.strhub_mark &&
( ./HipSTR \
--bams /data/in/input.bam \
--fasta /data/ref/hg38.fa \
--regions /data/in/regions.bed \
--str-vcf str_calls.vcf.gz \
); rc=$?; find . \
-xdev \
-type f \
-newer /tmp/.strhub_mark \
-size +0 \
! \
-path './.git/*' \
! \
-path './proc/*' \
! \
-path './sys/*' \
! \
-path './dev/*' \
! \
-path './data/*' \
2>/dev/null |
head \
-500 |
while IFS= read \
-r f; \
do mkdir \
-p "/data/out/$(dirname \
"$f")" &&
cp "$f" "/data/out/$f"; done; exit $rc
```

5. Reproducibility Metadata

Tool	HipSTR
Version	v0.7
Repository	https://github.com/tfwillems/HipSTR
Commit (pinned)	b2033bfbb5cf55496b776463bdf2993fa763a4be
Container OS	ubuntu-22.04
Marker panel	Autosomal STR
Verified on	2026-11-06 20:39:35 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/42289582063

Verified automatically, in a clean environment, on the tool's public source at the pinned commit. This certificate records what happened; it is not an endorsement by the tool's author.

6. Out of Scope

This report does not evaluate any of the following:

- Genotype correctness or accuracy
- Concordance against known truth sets
- Sensitivity, specificity, or stutter performance
- Allele calling accuracy or forensic casework suitability
- Regulatory compliance or ISO accreditation
- Multi-laboratory or multi-dataset reproducibility

7. Output Content Evidence

Output file	str_calls.vcf.gz
Format	VCF
Sequence records	21
Distinct STR loci	21
Total reads	7,851
Max depth (single locus)	517

8. Verification Data

The repository ships example data (3 file(s)); this run did not use it. STRhub ran the verification using the public reference dataset listed below. That dataset is a slice around 24 forensic STR loci, not a whole genome: it carries reads only at the loci listed below. A small test file in the repository lets a new user run the tool on their first day and see it working before trusting it with their own data, and it lets a verification run against the author's own sample as well as this one. Publishing the output that file should produce helps just as much: it shows what the results are meant to look like, which is what a reader needs to tell a correct run from one that merely finished.

Dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
Source	https://ftp-trace.ncbi.nlm.nih.gov/ReferenceSamples/giab/dat…
License	Open access (GIAB / NIST). Research and educational use. STRhub is not the data provider.
Ref. genome	GRCh38 / hg38
Loci tested	CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D18S51 · D19S433 · D1S1656 · D22S1045 · D2S441 · D3S1358 · D5S818 · D6S1043 · D7S820 · D8S1179 · FGA · PentaD · PentaE · TH01 · TPOX · vWA
Regions BED	Provided by STRhub

9. Verification Matrix

N/A	Run on repository test data	—
	Run on reference dataset	Illumina BAM (hg38), NA12878 (autosomal, female)
	README check (5/5)	Advisory: all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

10. What This Run Needed Beyond the Repository

The result above describes a run configured as follows. Anyone repeating it needs the same things.

– Test data: no sample from the repository was used, so a public reference sample stood in.
– A container environment, built from the tool's declared install steps rather than from a recipe the repository ships.

11. Notes from Reading the Repository

Recorded automatically from the tool's public files when this run was configured. Not verified by execution, and not part of the gates above. Useful for what to check by hand.

– Environment: generated by STRhub from the repository's Makefile (make), at the pinned commit.
– Input: the README documents BAM (line 69); this run used STRhub's illumina-bam-hg38 reference data.
– A comma-separated input list in the documented command was reduced to its first file.
– 1 input path(s) in the README command replaced with /data/in/input.bam.
– 1 reference FASTA path(s) replaced with /data/ref/hg38.fa.
– 1 BED path(s) replaced with /data/in/regions.bed.
– Run command: the README's own command, rewritten to STRhub's mounts; everything it created was captured as output.
– Regions: STRhub's ready-made hipstr file for the dataset's panel loci (STRhub records the hipstr layout for a program of this name).

12. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

13. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

14. Conclusion

- **Runs end-to-end**

  HipSTR v0.7 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit.

- **Produces plausible output**

  The tool generated a structurally valid VCF output file with genotype calls across 21 target forensic STR loci, with a maximum read depth of 517 at D18S51.

- **Verdict: Runs**

  The tool installed and its run produced its documented output.

- **Repository test data not used**

  The repository ships example data (3 file(s)), which this run did not use: the gates above ran on STRhub's reference dataset. STRhub Verified supplied Illumina BAM (hg38), NA12878 (autosomal, female) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

- **Reproducibility statement**

  The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix: Detected Markers with Read Depth

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LOCUS	READ DEPTH		LOCUS	READ DEPTH	
D18S51	517		**D6S1043**	360	
FGA	477		**D1S1656**	359	
vWA	456		**D12S391**	349	
PentaD	431		**D8S1179**	346	
D2S441	428		**D13S317**	316	
D3S1358	419		**D10S1248**	309	
D22S1045	414		**D19S433**	289	
D7S820	405		**TH01**	275	
PentaE	401		**CSF1PO**	275	
D5S818	399		**TPOX**	255	
D16S539	371				